

LOT Computer — Assembly Manual (Internal / Manufacturing Partner)

(Concept-stage — this is the DVT/PVT assembly reference to hand to whoever builds the units, per . . /06 -MANUFACTURING .md. Torque specs and exact fastener sizes are placeholders pending final DVT CAD.)

Exploded structure (front to back)

1 **Side B shell** (brushed stainless, front) — camera lens cutout, display window, button through-hole.

1 **Camera module** — seated and adhered into its cutout, ribbon cable routed to the PCB.

1 **Display module** — seated into its window, FPC connector routed to the PCB.

1 **Button cap** (stainless) — sits over the SMD tactile switch on the PCB, protrudes just enough through Side B's through-hole for a positive click.

1 **Main PCB assembly** — MCU, sensors, charge management IC, Qi receiver coil, battery connector, all reflow/placed by the SMT partner.

1 **Battery** — connected via JST, seated in its pocket, secured with adhesive foam tape (no rattling).

1 **Gasket** — seated in the mating groove between Side B and Side A.

2 **Side A shell** (polished stainless, back) — Qi contact patch aligned to the receiver coil beneath it.

1 **4× hex screws** — through Side A into Side B's threaded standoffs.

Assembly steps

1 Inspect Side B shell for cutout burrs; deburr if present before component seating — a burr near the display window can crack the FPC.

1 Seat camera module; connect ribbon cable to PCB header.

2 Seat display module; connect FPC to PCB header.

- 3 Place button cap.
- 4 Lower PCB assembly into Side B, aligning camera/display connectors and the button cap over its switch.

- 1 Connect battery JST; tuck cable away from the button mechanism.
- 2 Seat gasket into groove — check it sits flush, not twisted.
- 3 Lower Side A onto the assembly, Qi contact patch aligned over the coil.
- 4 Insert 4× hex screws, tighten in a cross pattern (not sequentially around

the perimeter) to seat the gasket evenly. Final torque spec: **TBD** — set once the fastener size is finalized in DVT; do not over-torque into stainless threads in the meantime.

- 1 Run the power-on self-test (see QA checklist, . . /06-MANUFACTURING.md §5).
- 2 Apply peel-off protective film to the polished Side A face.
- 3 Package.

Cautions

- The polished face scratches easily before the protective film is applied — handle by the edges/brushed side during assembly wherever possible.
- Do not force the gasket — if it doesn't seat flush, check for debris in the groove rather than compressing past it.
- Battery JST connector is polarized; do not force a reversed connection.